

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(d)	(i)	Any 2 × (1) from: - U235 fissile whereas U238 is not (1) - Only small amount (0.7 %) of U235 in natural uranium (1) - U238 absorbs neutrons	2			2		
		(ii)	Use of $v = r\omega$ to obtain $\omega = 2\,333\text{ [rad s}^{-1}\text{]}$ (1) $371\text{ [rev s}^{-1}\text{]}$ (1) Award 1 mark for answer of 186		2		2	2	
		(iii)	$0.7 \times 1.15^n = 5.0$ (1) $n = 14.1$ resulting in <u>15</u> stages required (1)		2		2	2	
			Question 10 total	6	9	5	20	10	0